EXPERIMENT 2

CO3 LAB-2
Day-2

Program:

```
b = [' '] * 9

def show():
    print(b[0:3])
    print(b[3:6])
    print(b[6:9])

for i in range(9):
    show()
    p = 'X' if i % 2 == 0 else 'O'
    n = int(input(p + " enter position (1-9): ")) - 1
    if b[n] == ' ':
        b[n] = p
    else:
        print("Position occupied!")
        continue

    if any(b[a] == b[c] == b[d] == p for a, c, d in
           [(0,1,2),(3,4,5),(6,7,8),(0,3,6),
            (1,4,7),(2,5,8),(0,4,8),(2,4,6)]):
        show()
        print(p, "wins!")
        break
else:
    show()
    print("Draw!")
```

Output:

```
[' ', ' ', ' ']
[' ', ' ', ' ']
[' ', ' ', ' ']
X enter position (1-9): 2
[' ', 'X', ' ']
[' ', ' ', ' ']
[' ', ' ', ' ']
O enter position (1-9): 9
[' ', 'X', ' ']
[' ', ' ', ' ']
[' ', ' ', 'O']
X enter position (1-9): 7
[' ', 'X', ' ']
[' ', ' ', ' ']
['X', ' ', 'O']
O enter position (1-9): 6
[' ', 'X', ' ']
[' ', ' ', 'O']
['X', ' ', 'O']
X enter position (1-9): 5
[' ', 'X', ' ']
[' ', 'X', 'O']
['X', ' ', 'O']
O enter position (1-9): 3
[' ', 'X', 'O']
[' ', 'X', 'O']
['X', ' ', 'O']
O wins!
```